

DATA VISUALIZATION · SUMMER 2026

120 Years of Olympic History

Who wins, and why?

A 12-question analytical journey through 120,000+ Olympic athlete records — gender progress, national strategy, and the physiology of a champion.

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1896–2016

Project Overview

120 years of Olympic athlete-event records (1896–2016), analyzed through 12 multi-dimensional questions spanning gender, nationality, physiology, and history.



Dataset

271,116 athlete-event records · 66 sports · 230 NOCs · Kaggle / sports-reference.com



Cleaning

Team-event medal rows deduplicated (one medal $\neq$ one row per roster athlete)



12 Analytical Questions

Each relates variables, compares groups, or tracks change — not single-column summaries



Visualization

Plotly (notebook) · CVD-safe palette · decluttered · titles that state the takeaway

The Dataset, At a Glance



104,061

Athletes



230

Countries (NOCs)



66

Sports



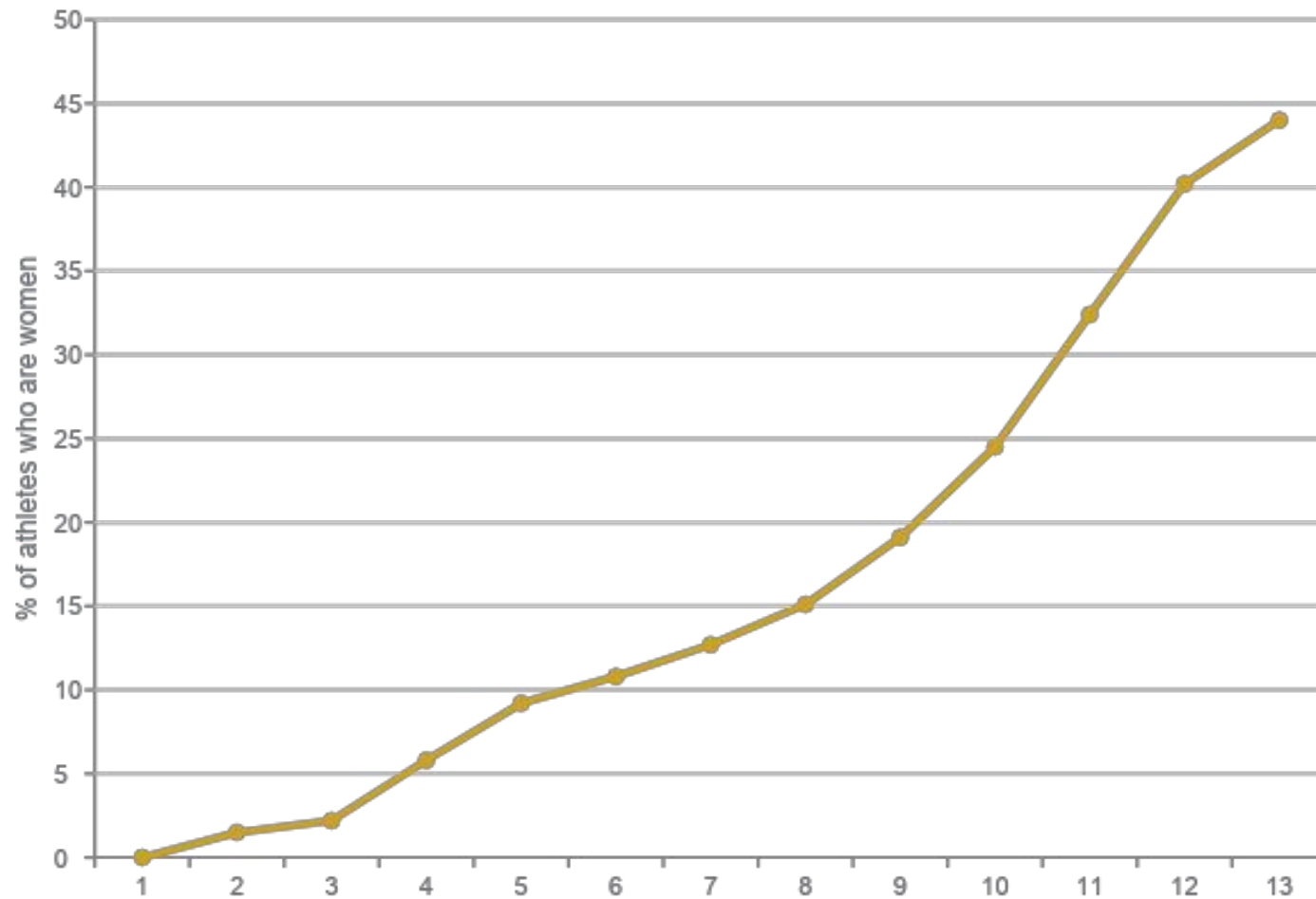
13,713

Medals awarded

Figures reflect a fixed 65% random sample of the full 271,116-row dataset (kept under GitHub's file-size limit) — sport, country, and year coverage are unchanged. “Countries (NOCs)” counts historical Olympic committee codes (e.g. USSR, East Germany are separate from Russia, Germany); the live dashboard reports 213 modern countries instead.

Q1

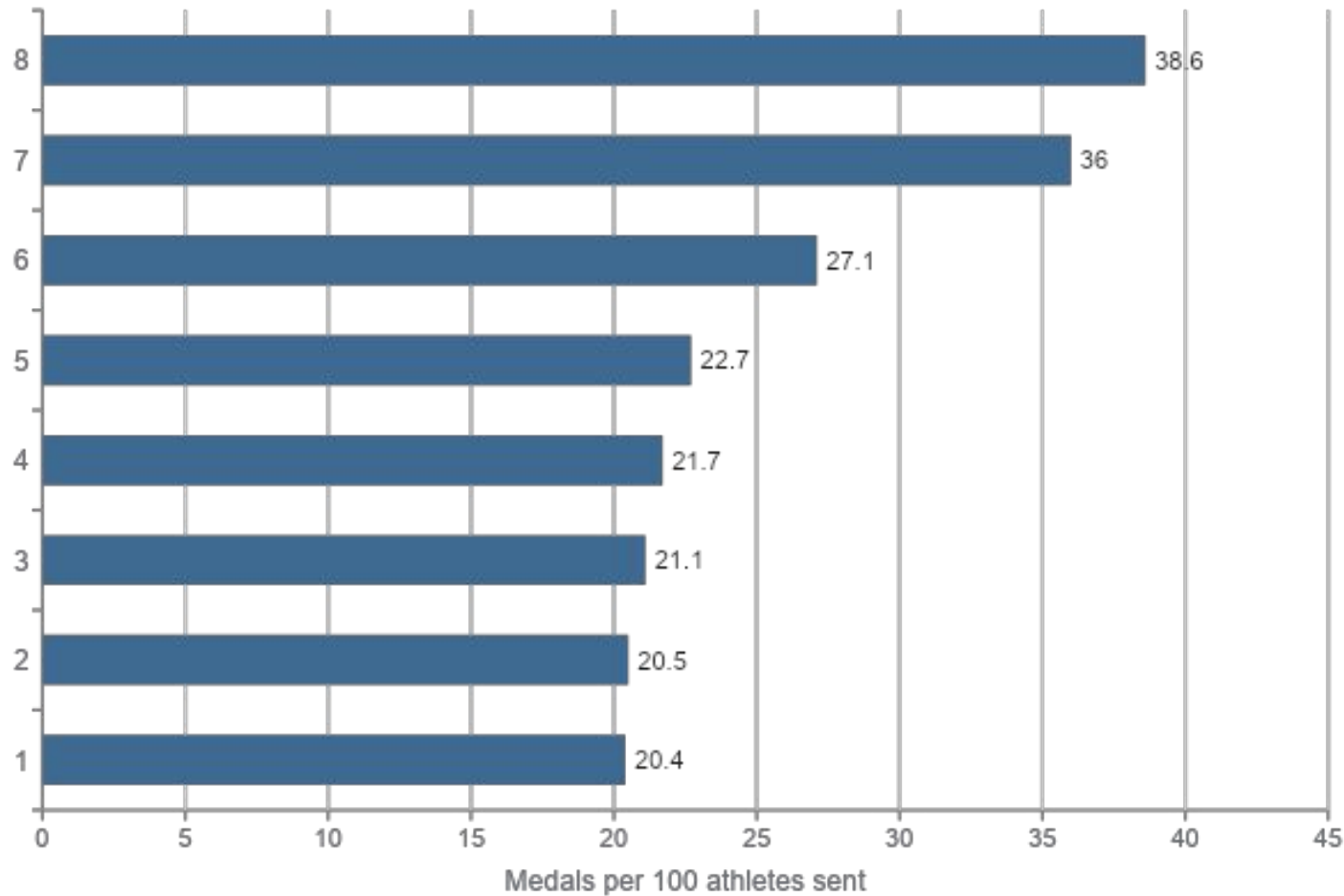
Women went from 0% to 44% of Olympic athletes over 120 years



💡 KEY INSIGHT

Female participation rose steadily, with the steepest gains after 1980 — the era of Title IX and IOC gender-equity pushes. The aggregate hides variation: some sports (boxing, wrestling) remain well below parity, while others (equestrian) are near 50%.

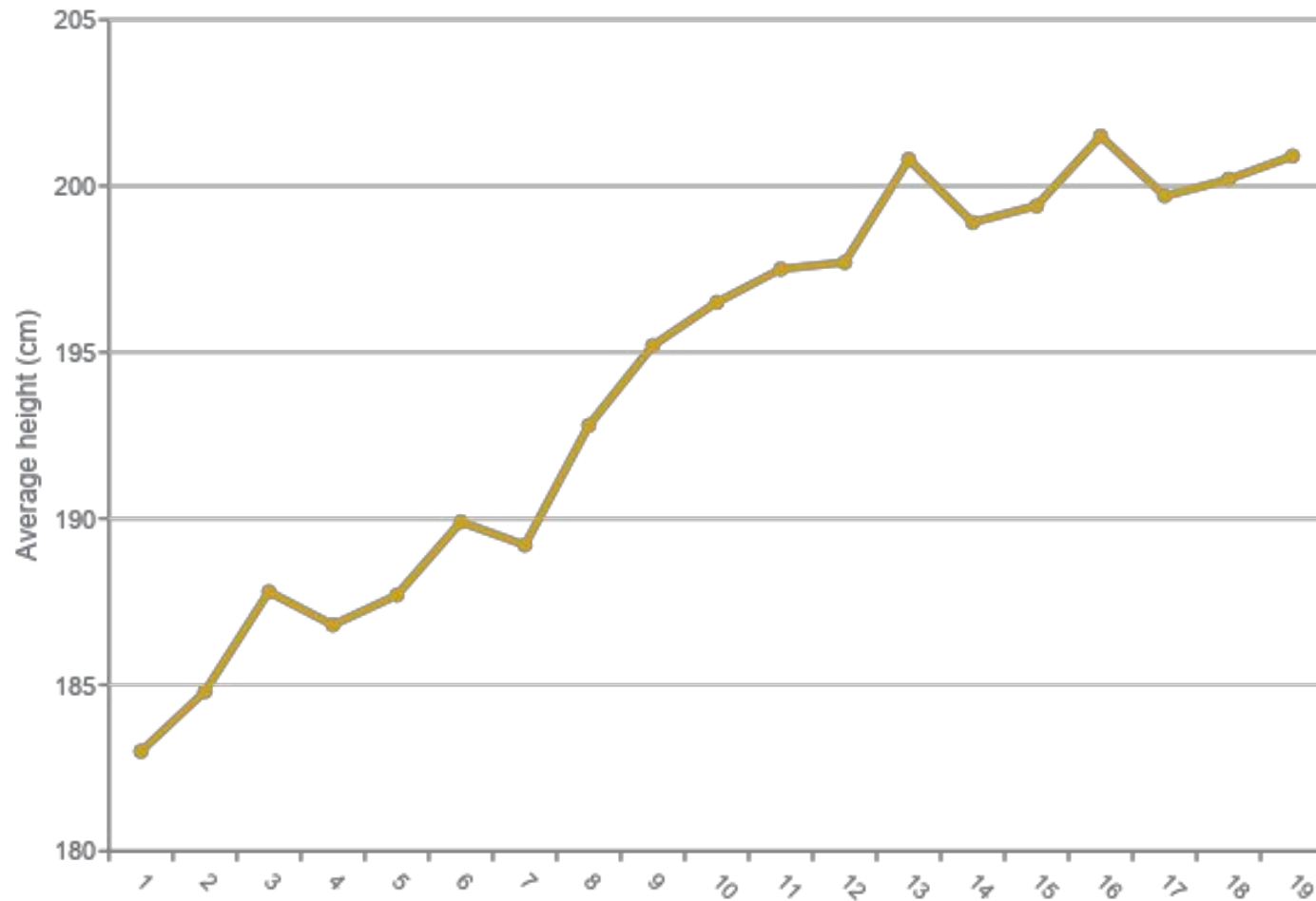
The USSR converted athletes into medals far more efficiently than the USA



KEY INSIGHT

Ranking by raw medal count just re-tells the story of national size. Normalizing by athletes sent reveals state-sponsored sport systems (USSR, East Germany) converted a far higher share of their delegation into medals — a different national strategy than the USA's broad-based approach.

Men's Olympic basketball players gained ~18cm in average height since 1936

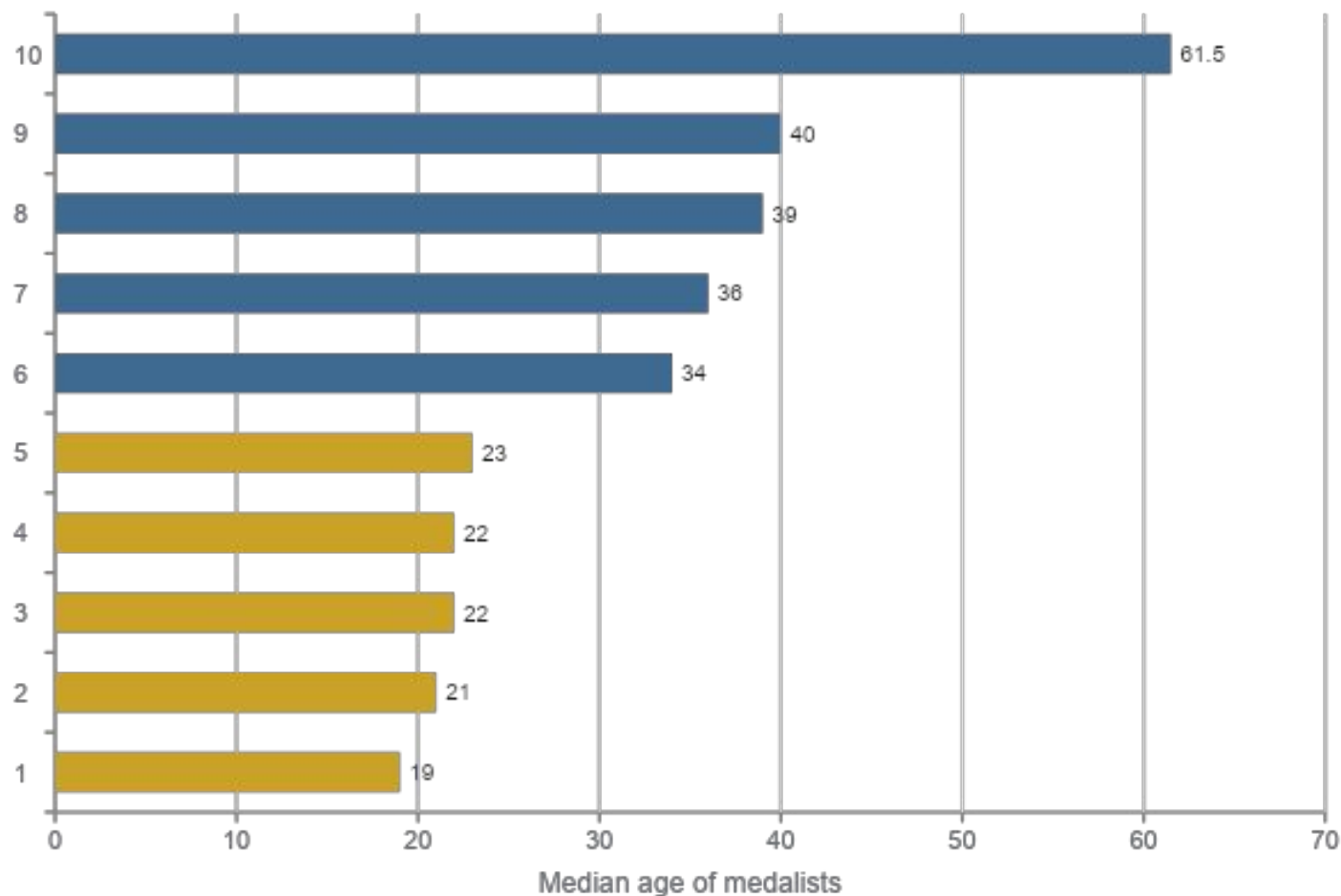


KEY INSIGHT

Elite sport increasingly selects for sport-specific physiques. Basketball's average height climbed steadily as the sport professionalized and scouting globalized — body type is a competitive filter, not an incidental trait.

Q4

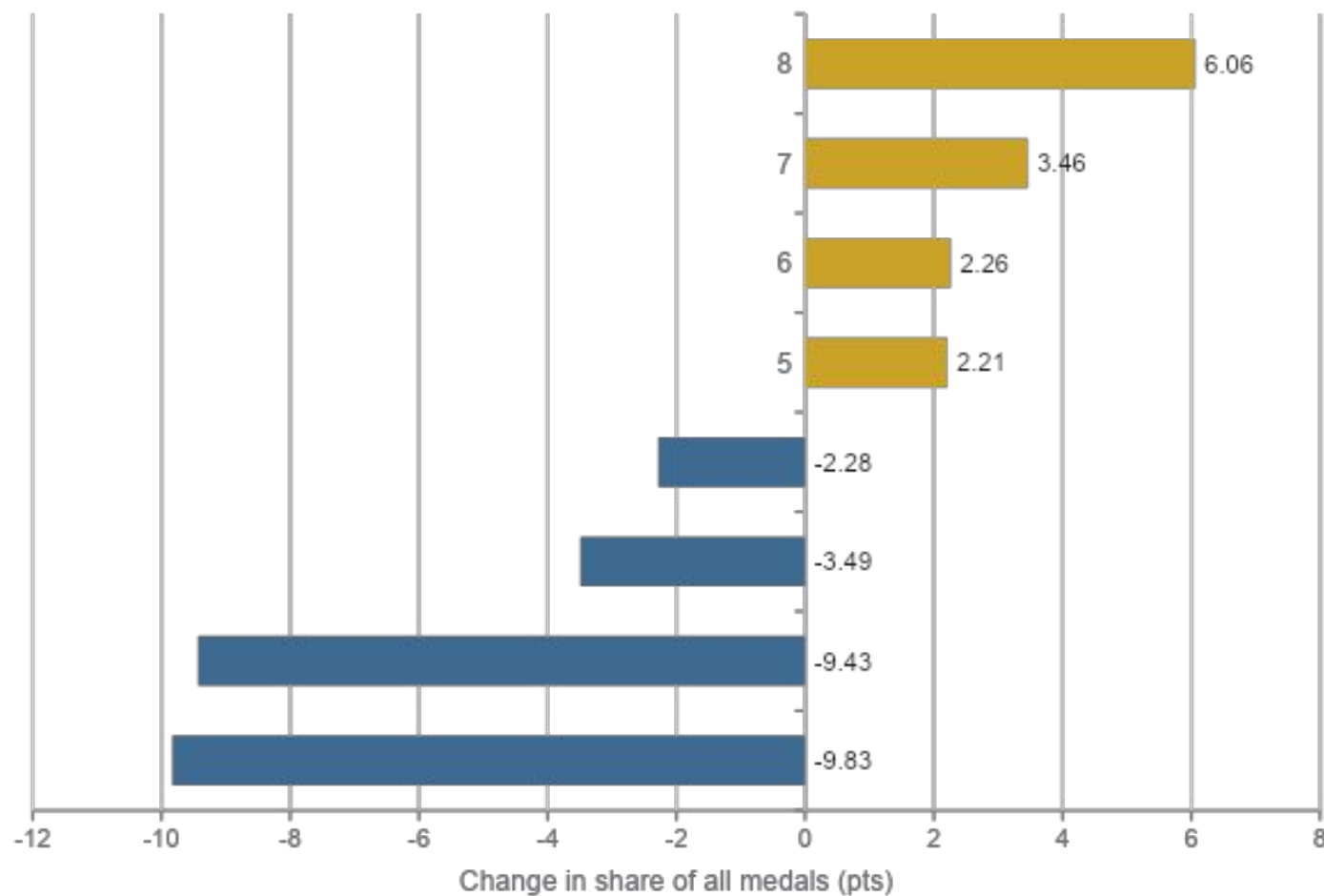
Medal-winning age spans three decades: teenage gymnasts to 60-something equestrians



KEY INSIGHT

“Peak age” is entirely sport-dependent. Power/flexibility sports judged on young, flexible bodies (gold bars) skew very young; precision and endurance sports where experience matters more than raw athleticism (blue bars) skew decades older — some medalists are in their 50s and 60s.

China's medal share rose 6 points since the 1970s; Germany's and Russia's fell most

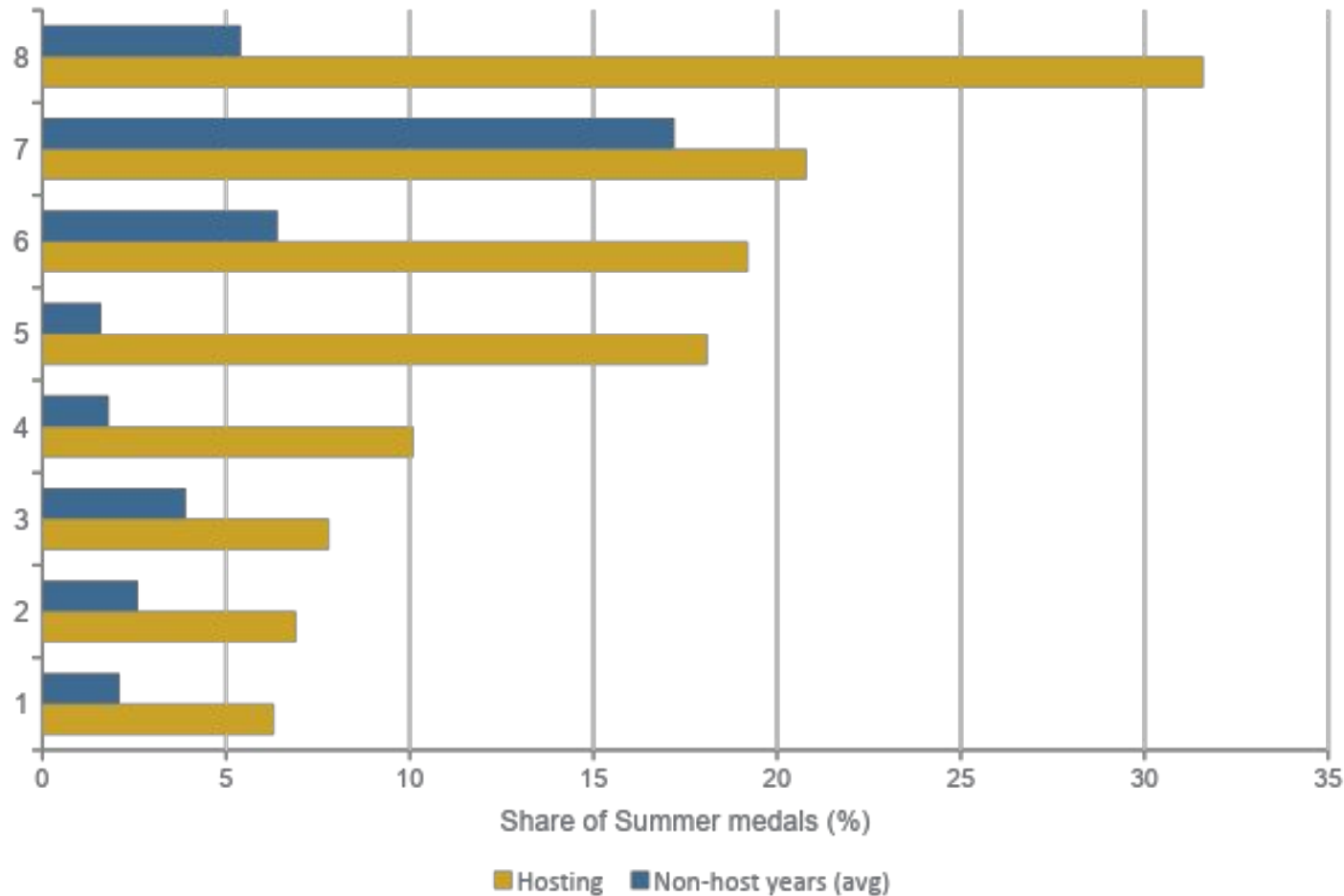


KEY INSIGHT

Comparing average medal share in the first vs. last 5 years of a 50-year window strips out single-Games noise. The result tracks real geopolitical shifts: the USSR/East Germany's collapse vs. China's sustained post-1980s investment in Olympic sport.

Q6

Hosting the Olympics reliably boosts a country's medal share

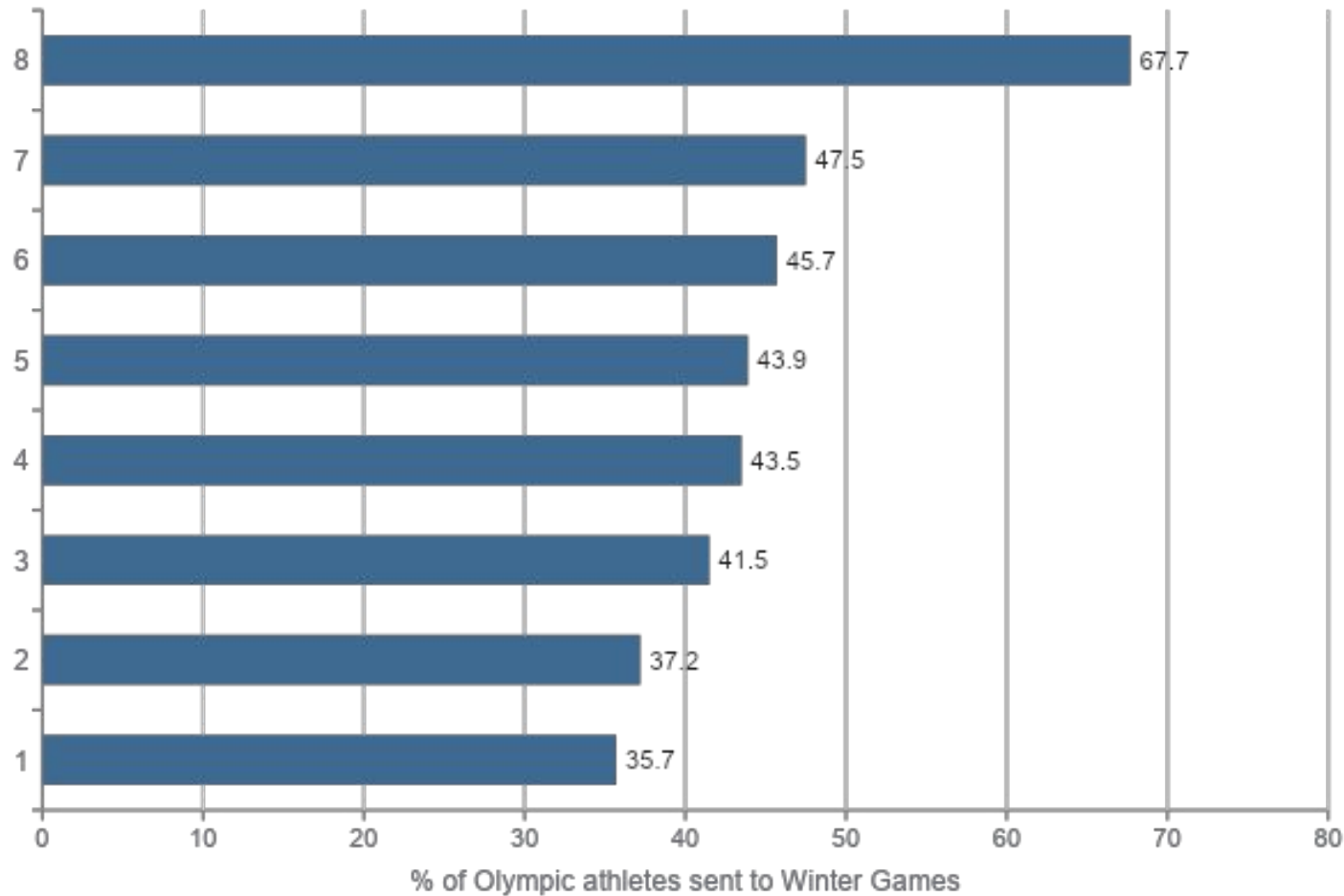


KEY INSIGHT

Nearly every host nation won a larger share of medals in their own Games than their historical average — consistent with a well-documented “home advantage” effect: funding cycles, automatic qualification, and a supportive crowd.

Q7

Alpine and Nordic nations send a far higher share of athletes to the Winter Games

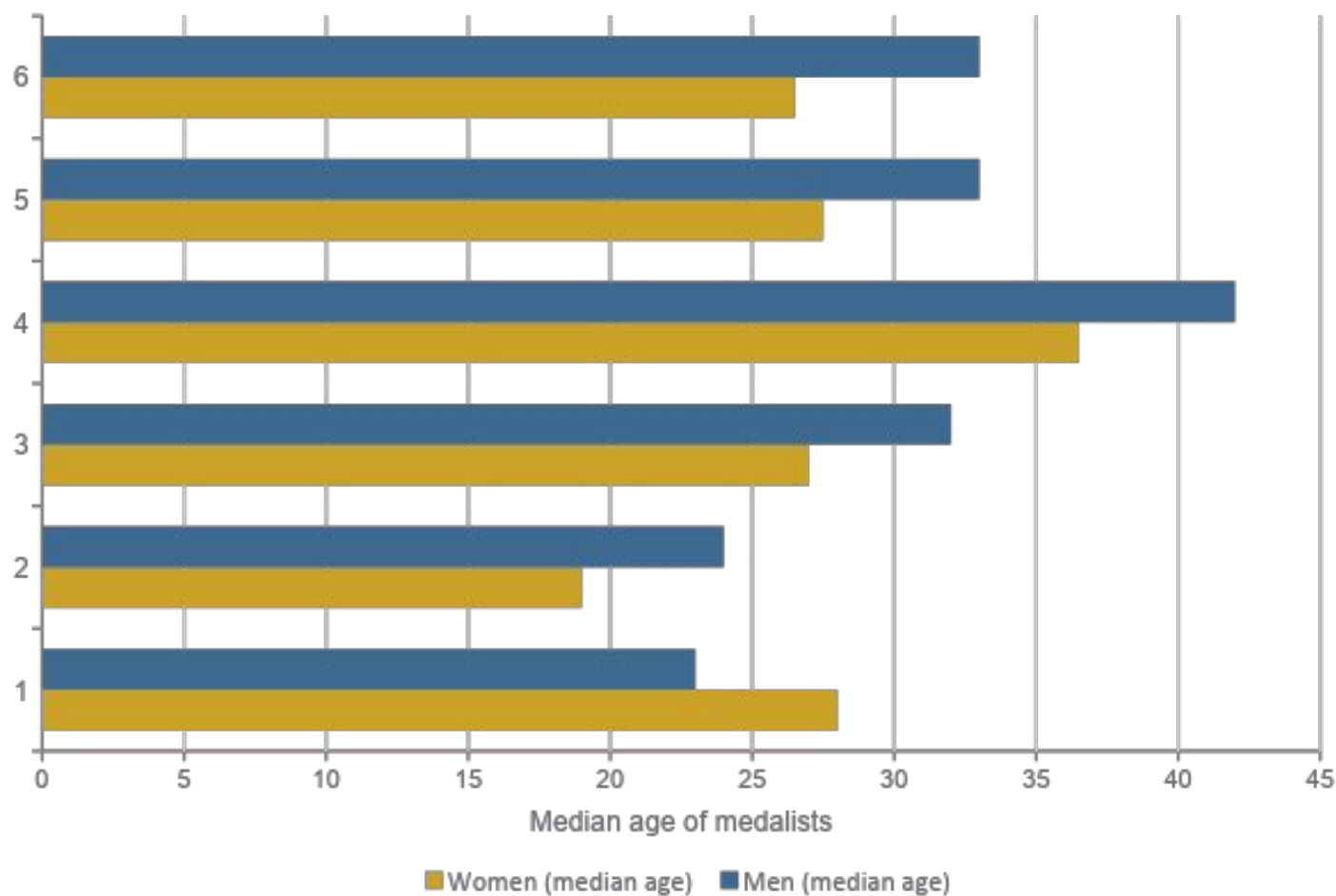


KEY INSIGHT

The ranking is a near-perfect proxy for winter-sport geography and climate: Alpine and Nordic nations send a much larger relative share of athletes to the Winter Games than warm-climate nations — participation infrastructure follows climate.

Q8

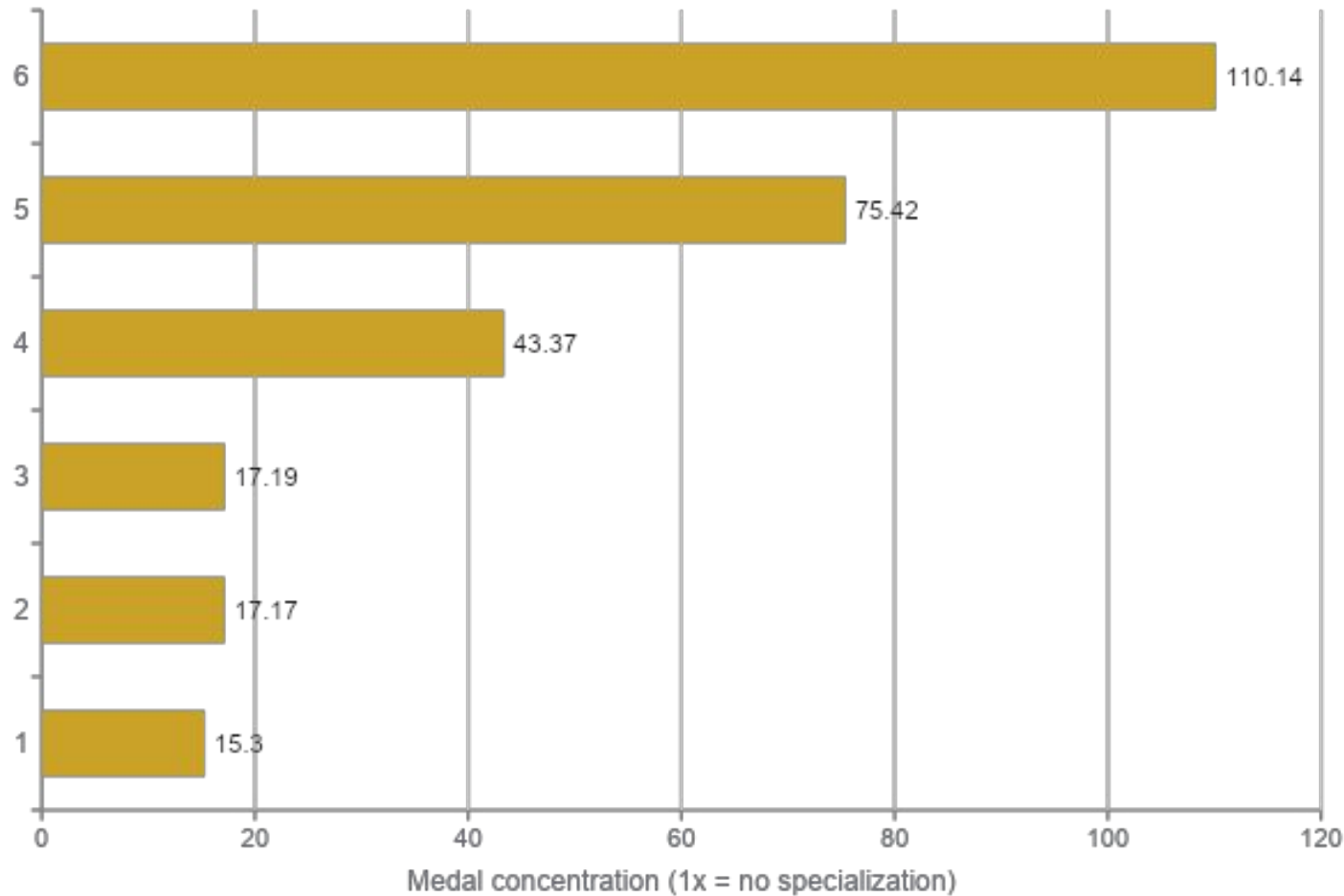
In several sports, men's medalists skew 5-6 years older than women's



KEY INSIGHT

Sports judged on youth and aesthetic execution (gymnastics) show little gap — both sexes peak young. In sports like shooting and sailing, men's medalists skew notably older, which may reflect historical differences in when women's events were introduced.

Some nations dramatically over-index in a single sport

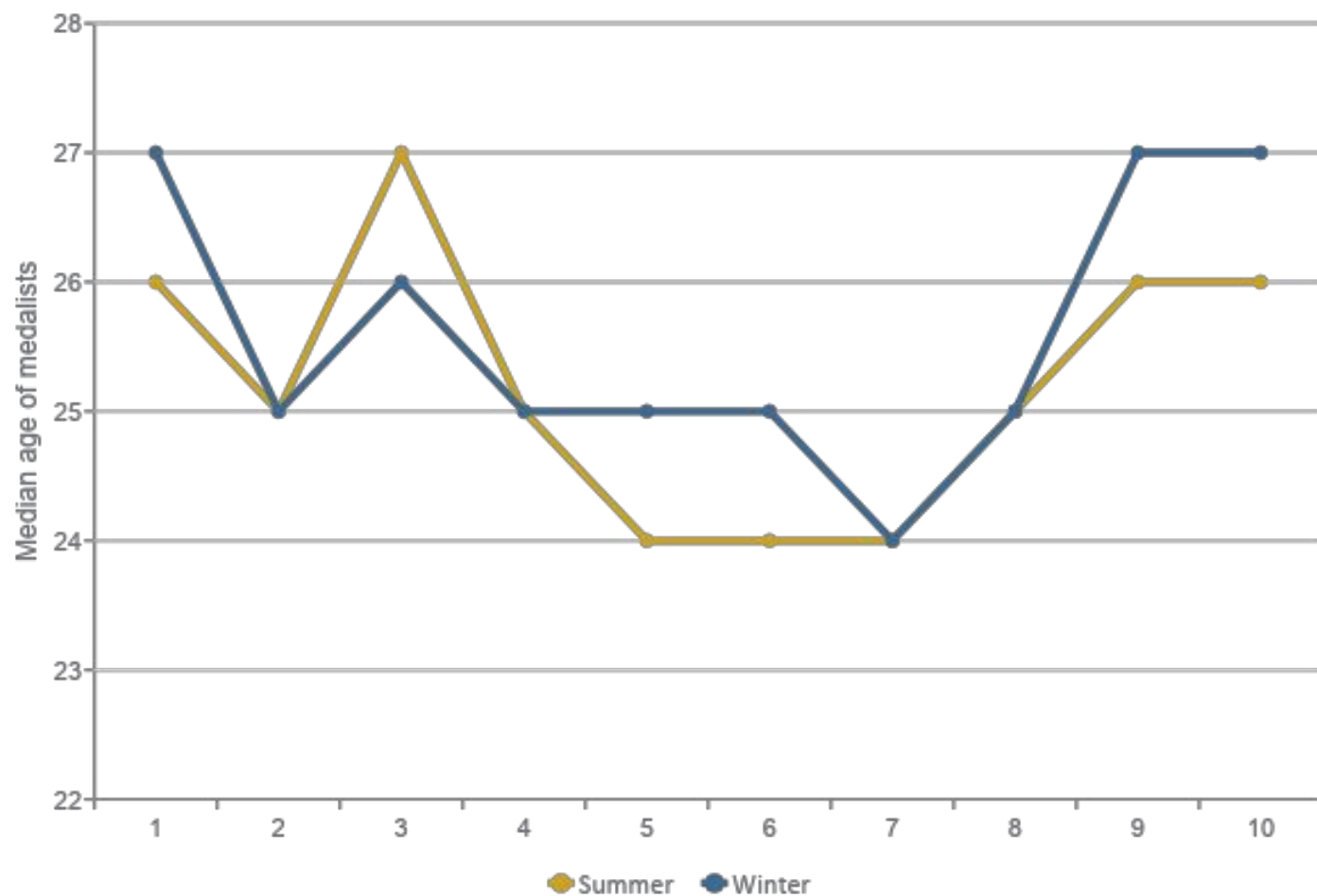


💡 KEY INSIGHT

A specialization index above 1 means a country wins a sport's medals far more than its overall profile predicts. Indonesia's badminton dominance (110x) and India's hockey legacy (75x) show small/mid-size nations can dominate within one sport — invisible on a medal-count leaderboard.

Q10

Median medalist age has crept up since the 1980s, in both Summer and Winter Games

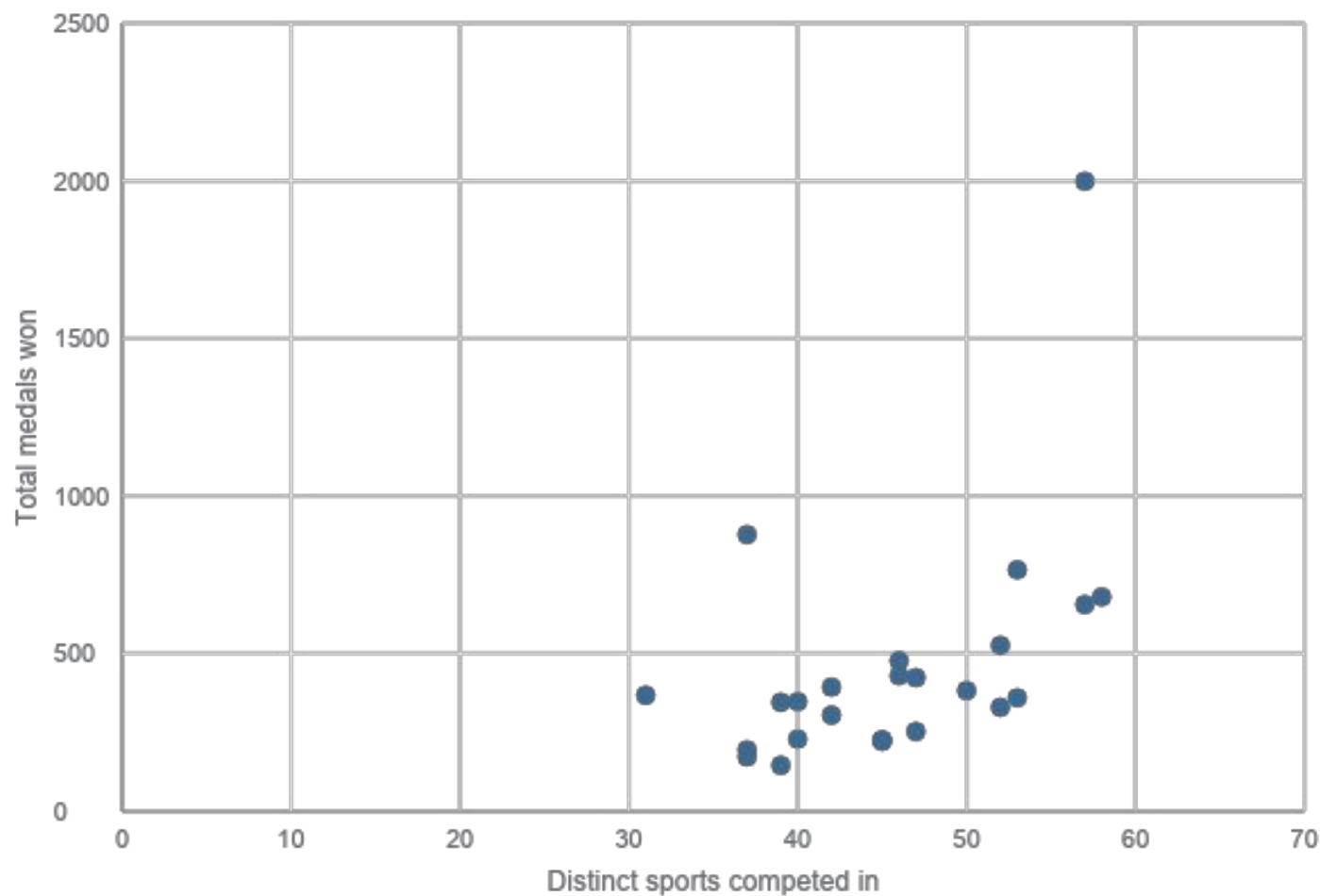


KEY INSIGHT

After a mid-century dip, medalist age has risen in both seasons since the 1980s — plausibly reflecting sports science, nutrition, and injury management extending elite careers, plus fewer one-off amateur competitors.

Q11

Broader sport programs win more medals overall ($r \approx 0.63$) — with big outliers

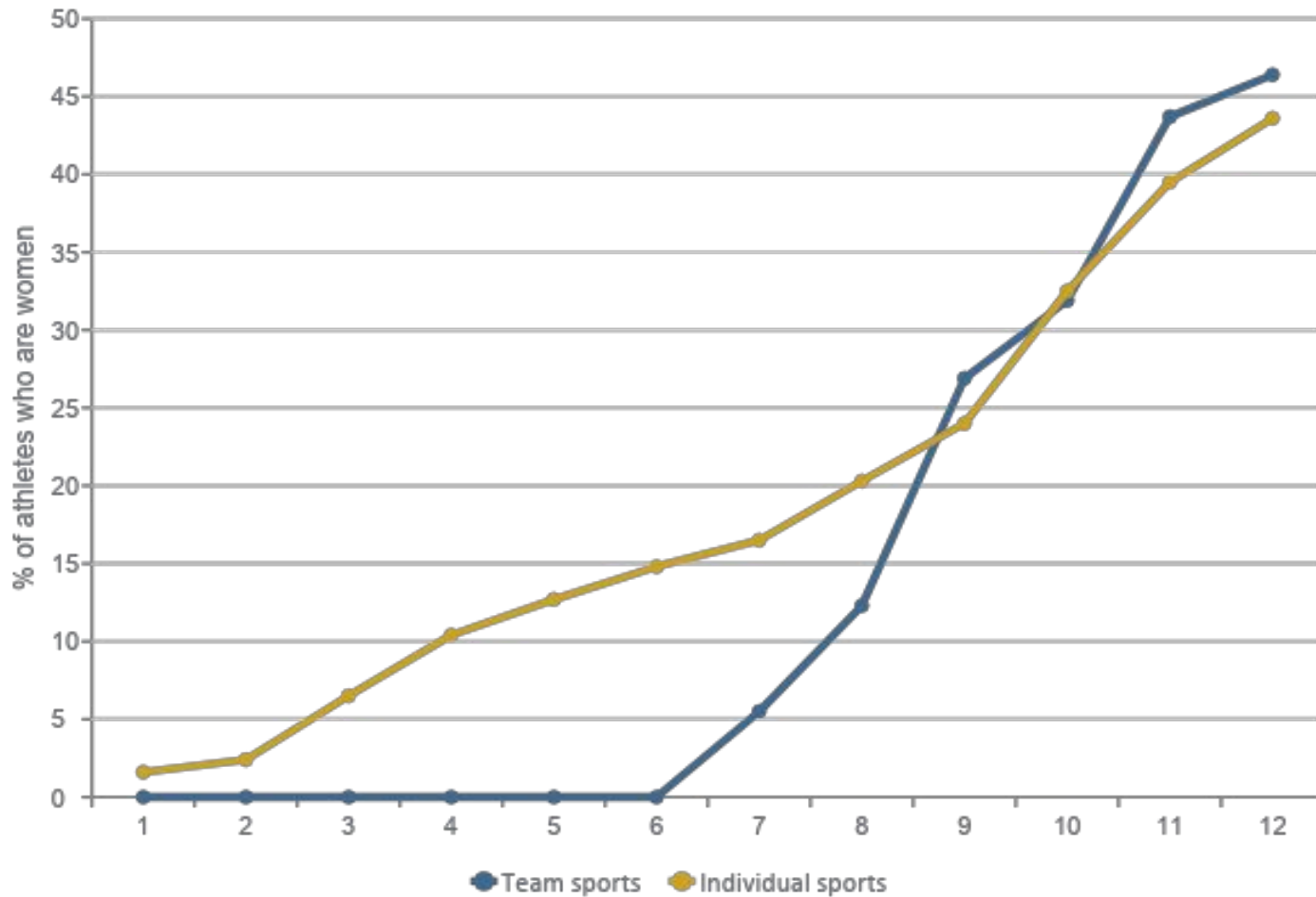


KEY INSIGHT

There's a strong positive relationship between how many sports a country competes in and total medals won — breadth is a reasonable proxy for Olympic investment. But specialist nations (see Q9) win far more than their narrow breadth would predict.

Q12

Women's participation grew in both team and individual sports, but team sports lag

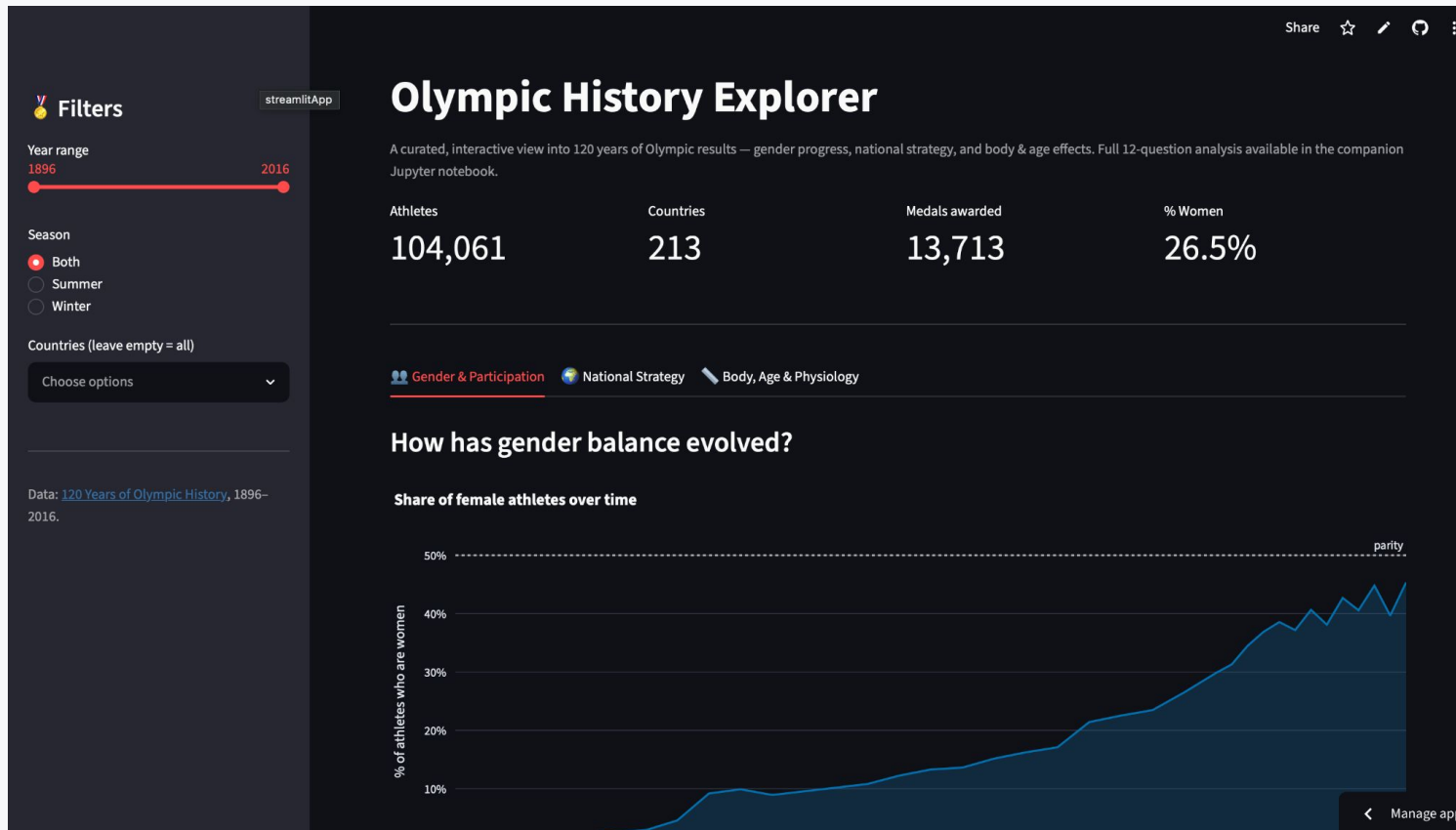


KEY INSIGHT

Team sports were introduced for women later and consistently trail individual sports in female participation share — team sports require entire new leagues and funding structures, which appears to slow gender-parity progress relative to individual events.

Explore It Yourself: The Live Dashboard

A curated, interactive subset of this analysis — built with Streamlit, deployed live from a public GitHub repo.



Gender & Participation

Female-share trend, sport breakdown by year, team vs. individual

National Strategy

Medal efficiency, specialization index, rising/declining nations

Body, Age & Physiology

Height by sport, youngest/oldest medalists, age by decade

Key Takeaways

- ✓ Gender parity is close in aggregate (44%) but structurally slower in team sports
- ✓ Efficient national strategy $\neq$ raw size — state-sponsored programs out-performed larger ones
- ✓ Hosting delivers a real, measurable medal-share advantage
- ✓ Body type and “peak age” are sport-specific and have shifted measurably over 120 years
- ✓ Specialist nations can dominate a single sport invisibly on a raw medal-count leaderboard

GITHUB REPOSITORY

github.com/NehalShivaram/Olympic_History_Analysis_Final_Project

LIVE DASHBOARD

olympichistoryanalysisfinalproject-krj2z7d9tnqnb8kgscv7rg.streamlit.app